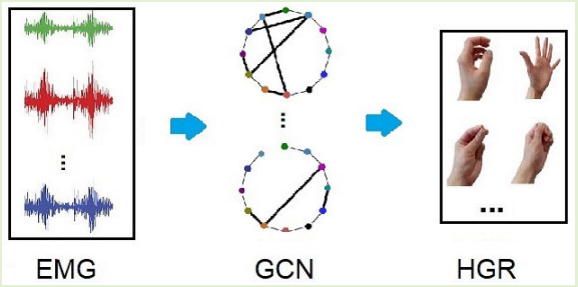
Hand Gesture Recognition From Surface Electromyography Signals With Graph Convolutional Network and Attention Mechanisms

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***Abstract*—In human body action recognition, graph convolutional networks (GCNs) show remarkable capability, compared to other deep learning (DL) methods, in capturing latent correlations within the human body topology. However, GCN methods have been rarely studied for hand gesture recognition (HGR) using surface electromyography (sEMG) because it is challenging to define a reliable topology across implicit muscle networks. In this article, we propose a novel covariance-based topology refinement module (CovTRM) to enable the GCN model to adaptively learn the dynamic topologies for various hand gestures. Extensive evaluations of two datasets, the Ninapro DB2 dataset and the UOW Dataset, show that the CovTRM can effectively refine the topologies to adapt to the implicit muscle synergies of different hand gestures. The proposed covariance-based graph convolutional network (CovGCN) model outperforms many machine learning (ML) models in recognizing sEMG-based hand gestures and mitigating the impact of variable limb positions, thereby contributing toward more effective and adaptable prosthetic hand control systems.**



***Index Terms*— Graph convolutional network, graph neural network (GNN), hand gesture recognition (HGR), surface electromyography (sEMG).**

I. Introduction

ELECTRIC hand prostheses have shown more advanced functionality and dexterity compared to their body-powered counterparts. As the cost and weight continuously decrease, electric hand prostheses have been growing rapidly in popularity among people with upper limb differences [1]. To control such a robotic hand, wearable sensors, typified by surface electromyography (sEMG), are attached to a user's residual arm and the recorded muscular bio-signals are used to activate the human–machine-interface (HMI). However, since its early introduction in the 1950s, this HMI has not progressed much in clinical practice, with the limited ON/OFF control of one preset hand gesture and requiring sequential manual switching for a new gesture. There is a high demand for electric hand prostheses that can be controlled intuitively and efficiently, which will significantly improve the quality-of-life of hand amputees and increase user-acceptance of the technology.

Hand movements involve complex muscle synergies, making it impractical to predict hand gestures by allocating muscle-specific sensors. Isolating bio-signals contributed from a certain muscle for a targeted degree of freedom (DOF) of hand is even a quite challenging task due to the existence of signal crosstalk. Therefore, machine learning (ML) is widely used to analyze the multichanneled bio-signals to identify muscle contraction patterns and map them to the intended hand gestures.

Early studies in sensor-based hand gesture recognition (HGR) have employed traditional ML classification

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algorithms, such as linear discriminant analysis (LDA), support vector machine (SVM), and random forest (RF). However, these ML techniques are not end-to-end but require feature extraction from raw signals beforehand. Finding general handcrafted features to proficiently discriminate the bio-signals led by different hand movements is quite challenging due to intersubject anatomical and physiological differences as well as the impacts of many confounding factors, e.g., variable limb position, muscle contraction intensity, electrode shifting, and concept drifting. Recently, with the capability of learning hierarchical representations and allowing the direct input of raw data, deep learning (DL) has been receiving tremendous popularity in the study of artificial intelligence (AI). A host of deep neural networks (DNNs) have been proposed, e.g., convolutional neural network (CNN), recurrent neural networks (RNNs), and deep belief network (DBN). The researchers are dedicated to extending the successes of these DNNs from computer vision (CV) and natural language processing (NLP) to other applications, including sensor-based HGR. The literature shows that the DL-based HGR systems can provide more robust HGR compared to the traditional ML-based counterparts [2].

Graph convolutional network (GCN) is a DL architecture that models the data as a structured graph. In the graph, a set of nodes is defined to represent some objects of interest, and the interdependencies between these variables are expressed as edges. GCN is good at modeling a context that contains a high number of spatial information with temporal correlations [3]. It has been widely studied in human body topology and action recognition and has generated outstanding achievements [4]. However, GCN is usually used for skeleton-based tasks and rarely for HGR based on sensors measuring underlying muscle activities [5]. One of the possible reasons is the difficulty of optimizing a graph with the presence of complex muscle synergies and signal noises [6].

In this study, we propose a covariance-based GCN architecture, using topology refinement graph convolution, to automatically optimize the graph and identify latent correlations between muscles to classify the intentions of hand gestures. Compared to other sensing modalities for muscles, e.g., mechanomyography (MMG) and force myography (FMG), sEMG is dominantly used as the bio-signal source in either unimodal or multimodal [7], [8] HGR systems since it generally contains richer neuromuscular information than the others [2]. Therefore, our proposed algorithm is tested and evaluated with the sEMG signals in this work. By comparing the results to those of other traditional ML methods (e.g., LDA, SVM, and RF) and DL methods such as CNN and Transformer, this novel GCN-based algorithm shows state-of-the-art performance in the classification accuracy of diverse hand gestures. It also performs better in dealing with the effect of limb position on HGR.

The main contributions of this article are as follows.

1) We introduce a new covariance-based topology refinement module (CovTRM), which is designed as a simple yet effective attention-based mechanism to capture the joint variability of sEMG channels. The module dynamically refines the prior topologies to produce channel-wise topologies, providing the pairwise relationships between the channels under different types of motion features.
2) We propose a new covariance-based graph convolutional network (CovGCN) for sEMG-based HGR. To the extent of our knowledge, this work is the first to represent sEMG signals with a dynamic graph. Instead of independently modeling the signals with a fixed topology, CovGCN automatically learns the channel-wise topologies in a refinement manner.
3) We provide extensive empirical results to validate the proposed method using the benchmark datasets. The experimental validation demonstrates that the proposed method improves the classification accuracy and the resilience to the impact of variable limb position (one of the major confounding factors affecting HGR accuracy). The results also show a favorable tradeoff between performance and model complexity.

The remainder of this article is organized as follows. Section II introduces the related work on automatic HGR using sEMG signals. Section III presents the proposed graph convolutional network for HGR. Section IV describes the datasets and preprocessing techniques used for our experiments, and provides the experimental results and analysis, and finally, Section V gives the concluding remarks.

II. Related Work

In this section, we provide a review of both the traditional methods and DL methods. The existing graph-based methods are then discussed.

A. Traditional ML Approaches for Sensor-Based HGR

Traditional ML approaches are still widely used for sensor-based HGR due to the relative simplicity and lower requirement on data amount. Generally, raw sensing signals cannot be directly used for HGR in traditional approaches. As a vital step, feature extraction need to be performed to capture useful representations from raw signals and use them as the inputs for the downstream classification or regression. Handcrafted features are typically used, and they can vary based on the types of wearable sensors. These features are often categorized into time domain, frequency domain, or time–frequency domain. The handcrafted features rely heavily on prior knowledge and have proven effective in HGR based on traditional ML techniques. Researchers often adopt a comprehensive approach by selecting a wide range of features and then applying feature selection tools to optimize the feature set. Three common feature selection methods are filter methods, wrapper methods, and embedded methods. The feature selection helps refine the feature set, leading to improved HGR outcomes.

When it comes to classification tasks for HGR, model training typically involves traditional ML approaches, including LDA, SVM, RF, ridge classifier (RC), multilayer perceptrons (MLPs), naive Bayes classifiers, decision trees, K-nearest neighbors (KNNs), and hidden Markov models (HMMs). There is no single model that can effectively tackle all tasks

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and datasets. To address this, researchers typically explore multiple algorithms and choose the most suitable one based on their specific needs, which may include accuracy and computational efficiency [9]. Furthermore, parameter tuning is a crucial step for the selected model(s) and optimizes the performance for the specific task at hand.

When dealing with dynamic gestures, dynamic time warping is commonly employed to identify the similarity between a template and a given sequence. This is necessary for dynamic gestures since the performing times and signal lengths vary. As a semi-unsupervised learning method, nonnegative matrix factorization (NMF) can be used to process sEMG signals for simultaneous and proportional control of prosthetic hands. The application of NMF to sEMG-based HGR is motivated by the theory of muscle synergy theory, which aims to understand how muscle networks operate to generate hand/arm movement.

B. DL Approaches for Sensor-Based HGR

As wearable technology continues to advance, the difficulty of acquiring abundant HGR data is dramatically decreasing. Many good-quality databases in this field are publicized [2]. This significantly enriches the data availability and consequently promotes the rapid development of DL in this research area. The bio-signals for HGR are essentially time-series signals. Therefore, numerous RNN algorithms such as long short-term memory (LSTM) are widely used to extract temporal features due to their good performance in sequence-to-sequence (seq-2-seq) tasks [10]. The CNN has also shown its capability in temporal featurization by performing convolutions for a sequence of data from each sensor [11], [12]. Besides, the CNN's advantage in extracting spatial features makes it a strong alternative to learning the synergies across various muscles reflected by the multichannel bio-signals. Hence, the CNN is the most used DL technique due to its efficacy in the extraction of both temporal and spatial features. It can be used by itself or sometimes as a part of hybrid architectures such as CNN-RNN [13]. Recently, the popularity of Transformer based on attention mechanisms has been soaring with outstanding successes in both CV and NLP. A noticeable work [14] using Transformer for HGR is reported showing the results of state-of-the-art classification performance compared with some other ML methods.

C. Graph Convolutional Networks

Graph neural network (GNN) is generally employed to process non-Euclidean data such as graphs. As the most used GNN, GCN has two main categories: spectral approaches and spatial approaches. The former category focuses on tasks that convolution can only be conducted in the spectral domain while the latter can conduct convolutions directly on the defined graphs with predetermined feature update rules. Our proposed model is a spatial approach, adopting a two-step feature update rule reported in [15].

The key challenge in GCN is topology modeling, which is finding the best topology that reveals the relationships among all the nodes. According to the types of topologies, GCN can be categorized as static methods and dynamic methods. In a static method, the topology is predefined and does not change in the modeling. Whereas the topology can be adjusted and updated in a dynamic method. With a dynamic topology, GCN can be more general when facing some tasks without clear knowledge about the relationships between the nodes [4]. Dynamic methods are widely used for action recognition tasks.

There are frequent applications of GCN for skeleton-based motion tracking or recognition tasks, in which the connections of nodes are usually explicit and it is relatively simple to define the topology. However, there is limited literature about using GNN or GCN for sEMG-based HGR. Massa et al. [3] reported the first attempt using GNN to process the data collected by two high-density (HD) sEMG (64 channels for each) sensors to recognize 65 gestures. The overall error rate was 8.75%. However, there was no comparison with other ML/DL methods. In another work, Lai et al. [6] proposed the spatial–temporal convolutional networks (STCNs) and tested the regular GCN on three publicly available sEMG datasets (CapgMyo DB-a, CapgMyo DB-b, and BandMyo), two of which are HD sEMG datasets and the other is a sparse one (recorded by Myo Armband). By comparing the performance using the same datasets, the authors showed the superiority of the GCN over the other ML/DL methods in capturing spatial information. Lee et al. [5] developed an 8-channel stretchable sEMG array, and used the self-attention integrated GNN model to classify 14 static and four dynamic gestures. Their method outperformed the baseline method (STCN model in [6]) by approximately 4% in classification accuracy. Zhong et al. [16] proposed a spatio-temporal GCN for gesture recognition using a publicly available HD sEMG dataset with 65 gestures. Their method achieved higher classification accuracy than an LSTM model (91.07% versus 83.3%). Research on GNN/GCN-based HGR is limited, with most studies focusing on HD sEMG data. One major reason is the challenge of modeling implicit relationships among the nodes, i.e., the sEMG signals and their underlying muscles. Hand gestures are generated by complex muscle synergies. Furthermore, there are signal crosstalks and signal noise making it challenging to identify a satisfactory topology for a GNN or GCN for sEMG-based HGR. For prosthetic hand control, sparse sEMG is often considered more practical than HD sEMG due to its ease of use and lower hardware complexity. However, studies with sparse sEMG for GNN/GCN-based HGR are even rarer, highlighting a significant gap in the literature.

D. Impact of Variable Limb Position for Sensor-Based HGR

Considerable research has been undertaken in controlled environments to enhance the consistency of bio-signals and improve HGR outcomes [2]. As a result, such sensor-based HGR systems typically exhibit superior performance under optimized conditions. However, a significant challenge in prosthetic hand control, known as the "limb position impact," presents a notable obstacle that must be addressed [17]. This impact can degrade the HGR robustness and the performance of prosthetic control consequently [18]. Therefore, achieving precise gesture recognition across different limb positions is crucial for ensuring the viability of prosthetic control in

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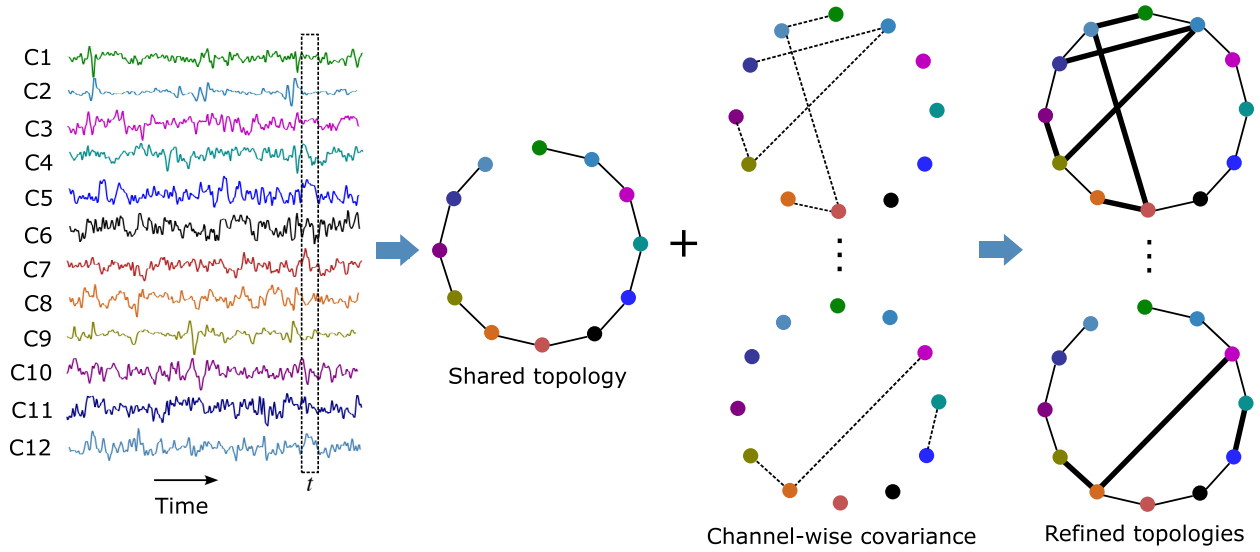


Fig. 1. Examples of sEMG signals (left) and graph representation with channel-wise topology refinement (right). The thickness of the lines indicates the relationship strength between the sEMG channels. Best viewed in color.

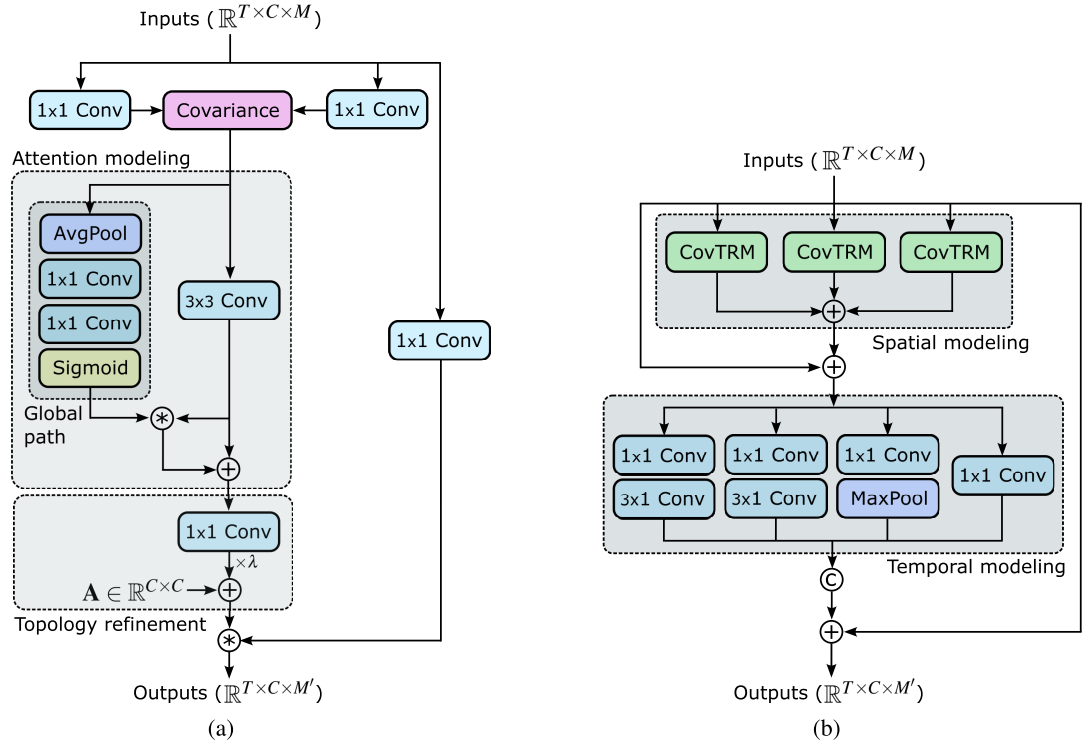


Fig. 2. Proposed modules of CovGCN. For brevity, "*Conv*" stands for convolution; "*MaxPool*" stands for max pooling; and "*AvgPool*" stands for average pooling. (a) CovTRM. (b) GCB.

practice. Researchers have increasingly focused on improving performance by gathering gesture data from various limb positions, a strategy shown to enhance overall system effectiveness [8], [17], [19], [20], [21], [22]. The literature also reveals that the DL approaches (typically based on CNN/RNN models) usually outperform traditional ML approaches (e.g., LDA, RF, and SVM) in overcoming the impact of limb position on classification accuracy. However, research addressing limb position's effects on HGR remains limited. This calls for more investigation on new hardware-based or algorithm-based approaches that are more resilient to such confounding factors.

III. Proposed Graph Convolutional Network

This section presents the proposed CovGCN, including graph representation (see Section III-A), CovTRM (see Section III-B), and graph convolutional block (GCB) (see Section III-C).

A. Graph Representation

Let $\mathbf{S} \in \mathbb{R}^{T\times C}$ be a segment of sEMG signals, where T is the temporal dimension and C is the number of sEMG channels (i.e., sensors). The signal data at the tth time step can be represented as a graph with the channels as vertices

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and sequential connections between the channels as edges (see Fig. 1). Formally, the graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ includes the vertex set $\mathcal{V} = \{v_i | i = 1, 2, \ldots, C\}$ and the edge set $\mathcal{E}$ which is denoted as an adjacency matrix $\mathbf{A} \in \mathbb{R}^{C \times C}$.

The adjacency matrix can be utilized as a shared topological prior that provides initial correlations between the sEMG channels. Let $\mathbf{x}_i \in \mathbb{R}^M$ be the feature vector of the ith channel, where M denotes the feature dimension. The topology-shared graph convolution updates this representation using a linear transformation and feature aggregation

$$\mathbf{x}'_i = \sum_{v_j \in \mathcal{N}(v_i)} a_{ij} \mathbf{x}_j \mathbf{W} + b_i \tag{1}$$

where a_{ij} is an element in the adjacency matrix, $\mathbf{W}$ is the weight matrix, $\mathcal{N}(v_i)$ is the neighbors of v_i, and b_i is the bias term.

Inspired by Chen et al. [4], the proposed CovGCN dynamically refines the shared topology with channel-wise covariance during the training process, which allows for automatically capturing subtle relationships between the sEMG channels. This refinement method avoids modeling the signals independently with a fixed topology as in the existing graph-based methods [3], [5]. The graph representation is illustrated in Fig. 1, together with some examples of multichanneled sEMG signals.

B. Covariance-Based Topology Refinement Module

Based on the aforementioned graph representation, we propose the CovTRM as shown in Fig. 2(a). Given an input feature $\mathbf{X} \in \mathbb{R}^{T \times C \times M}$, the CovTRM first performs two 1×1 convolutions as linear transformations for dimensionality reduction. The channel-wise covariance representing the relationships between two feature vectors $\mathbf{x}_i$ and $\mathbf{x}_j$ of the sEMG channels is then given by

$$\text{cov}(\mathbf{x}_i, \mathbf{x}_j) = \mathbb{E}[(\mathbf{x}_i - \mathbb{E}[\mathbf{x}_i])(\mathbf{x}_j - \mathbb{E}[\mathbf{x}_j])]. \tag{2}$$

To enhance the most relevant parts in the covariance matrix, we design a simple yet effective attention mechanism with a global path and a 3×3 convolution as the residual connection. The global path consists of an average pooling operation, followed by two 1×1 convolutions with the sigmoid activation function. The attention maps obtained by this path can be utilized as the global semantic information for refining the input covariance.

Let $\mathbf{V} \in \mathbb{R}^{C \times C \times M'}$ be the learned channel-wise covariance produced by the attention module. The refined topologies $\mathbf{R} \in \mathbb{R}^{C \times C \times M'}$ can be obtained by refining the shared topology $\mathbf{A}$ as follows:

$$\mathbf{R} = \mathcal{R}(\mathbf{V}, \mathbf{A}) = \mathbf{A} + \lambda \mathbf{V}. \tag{3}$$

Here, the refinement weight λ reflects the tradeoff between the initial correlations and the learned channel-wise relationships of the channels. It is a trainable weight (not a hyper-parameter) that is automatically adjusted during training to minimize the loss value. Finally, a batch matrix multiplication is used to combine the input features with the refined topologies.

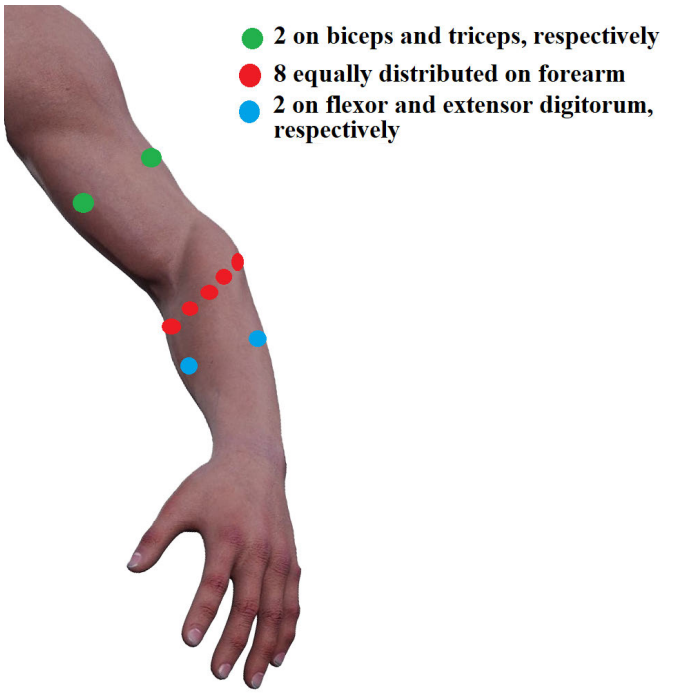


Fig. 3. Placement of 12 sEMG sensors from Ninapro DB2.

C. Graph Convolutional Block

Based on the proposed CovTRM, we introduce a GCB as shown in Fig. 2(b). Inspired by Chen et al. [4], the GCB consists of two sequential modules to process spatial and temporal features separately. The spatial modeling aims to capture intrinsic correlations between the sEMG channels within a time step. To this end, we utilize three CovTRMs in parallel to refine the self-link topology, inward topology, and outward topology, respectively. The outputs from these three branches are summed up before applying batch normalization and the ReLU activation function.

The temporal modeling aims to extract features at multiple scales along the temporal dimension. This module has four convolutional branches, which are designed based on [23]. The first two branches consist of two consecutive 1-D convolutions with a kernel size of 1×1 and 3×1, respectively, which allow capturing temporal patterns. To enhance both local and global semantic information, we increase the receptive fields of the 3×1 convolutions by utilizing different dilation rates. Finally, the extracted temporal features from the four branches are concatenated to obtain the final outputs.

Based on the proposed GCB, we construct a simple yet powerful graph network for sEMG-based hand gesture recognition (HGR), called CovGCN. It is designed as a sequential model with L consecutive GCBs, followed by a global average pooling, a dropout layer, and a softmax classifier. The output feature dimension M' is doubled after every two GCBs.

IV. Experiments and Analysis

A. Datasets for Experiments

1) NinaPro DB2 Dataset: Numerous newly proposed HGR algorithms were tested and evaluated on some publicly available datasets, which not only reduces workloads in collecting

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data but also makes it simpler to compare the performance across different studies. Ninapro [24] is one of the largest online sites providing open-source datasets for the research in HGR with various wearable sensors. In this study, we used the labeled sEMG data from the second database (referred to as DB2) of Ninapro for training and validation of the proposed CovGCN model. The data was collected from 40 intact subjects (11 females and 29 males, aged from 23 to 45). Twelve Delsys Trigno sensors were attached to each participant's arm to measure and record sEMG signals with a sampling rate of 2000 Hz. The detailed sensor location is shown in Fig. 3.

Three sets of exercises (B, C, and D) are conducted by each subject [25]. In Exercise B, 17 hand/wrist gestures with a bare hand are repeated six times with each gesture lasting for 5 s every time. Exercise C includes 23 hand grasps with some common objects. Exercise D includes nine gestures, which are the flexion/abduction of a single finger or two fingers. Therefore, there are totally 49 hand/wrist gestures (excluding "rest") in the whole database. Consistent with some reported studies [14], [26], [27], [28], [29] for comparison, four (1, 3, 4, and 6) out of six repetitions are employed as the training dataset while the other two repetitions are employed as the testing dataset.

Several preprocessing steps (e.g., filtering and segmentation) are usually required before inputting the sEMG data to a neural network. The data retrieved from Ninapro DB2 was already pro-processed by a Hampel filter to remove the power-line interference at 50 Hz from the raw signals. No low-pass filter was applied in the present study as high-frequency EMG information may also contribute to the discrimination of different muscle contractions. For the HGR applications, an acceptable time latency was suggested to be less than 300 ms with the optimal range between 100 and 125 ms [30]. Therefore, in our study, the data was segmented into sliding windows of 200 ms (400 samples) with a stride of 50 ms (100 samples) for each EMG channel/sensor.

2) UOW Dataset: In our previous work [8], we employed a custom-made armband equipped with eight sEMG sensors to record data across diverse hand gestures and various arm positions, aiming to evaluate the efficacy of the proposed CNN model in enhancing HGR performance, particularly considering the impact of limb position in practical scenarios. This study involved ten participants (seven females and three males, all right-hand dominant). Ethical approval for the study was obtained from the Human Research Ethics Committee at the University of Wollongong. We accumulated all the collected data into one dataset and named it "UOW Dataset."

In the experiment, the participants wore the sEMG sensors equally distributed around the forearm, following clear instructions with voice cues to perform specific arm positions and gestures. Data acquisition was performed using a Teensy 4.1 microcontroller board at a high sampling rate of 2000 Hz. Participants presented each three arm positions [see Fig. 4(a)] with nine different hand gestures [see Fig. 4(b)]. The whole experiment included two rounds: one for training and the other for testing, with each gesture repeated 12 times across varying arm positions. Each gesture was held for one second with a 3 s rest in between to ensure presenting the accurate gesture, and a 5 min break was given between two rounds to prevent fatigue.

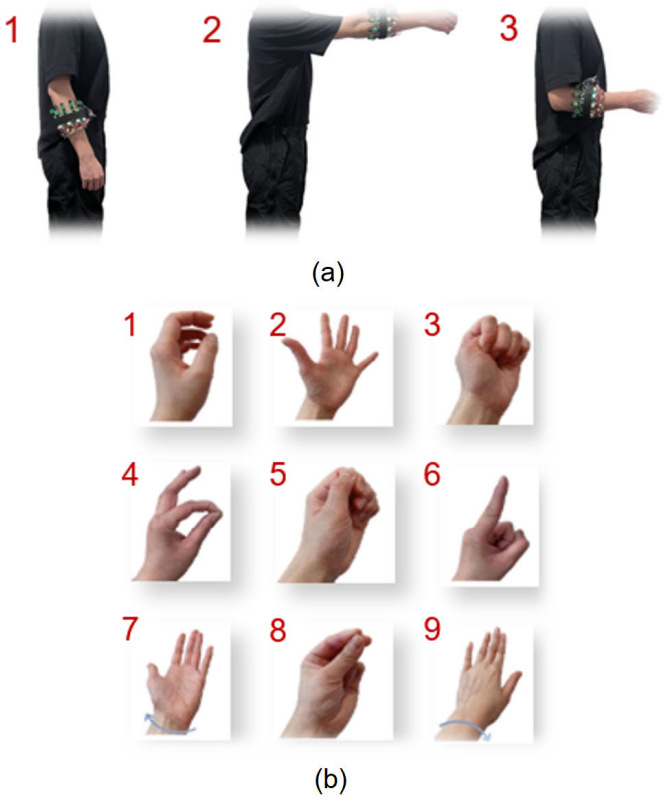


Fig. 4. Presented arm positions and hand gestures. (a) Three arm positions [1: relaxation at side; 2: 90° pointing forward; 3: 90° elbow flexion]. (b) Nine hand gestures [1: relax; 2: open; 3: fist; 4: pinch; 5: key; 6: point; 7: pronation; 8: tripod; 9: supination].

For the data recorded with our 8-channel sEMG armband, we processed the EMG signals using a series of filters: a low-pass filter (150 Hz cut-off) to remove high-frequency noise, a high-pass filter (20 Hz cut-off) to eliminate low-frequency artifacts, and a notch filter to eliminate 50 Hz power line interference. To directly compare with the results from our previous study using the same armband with CNN model [8], we maintained consistent setups. The data was segmented into sliding windows of 200 ms (400 samples) with a stride of 20 ms (40 samples) for each EMG channel/sensor.

B. Implementation Details

All our modeling works are performed by using an NVIDIA GeForce RTX 4090 24-GB GPU with the PyTorch DL framework. The models are trained with the Adam optimization algorithm with a learning rate of 0.001. The batch size is 64 and the number of epochs is set as 200 for training.

C. Learned Graphs of Muscle Synergies

Fig. 5 illustrates the learned adjacency matrices by the CovTRM-based model for one subject from the NinaPro DB2 dataset. The magnitudes are scaled from 0 to 1, and each adjacency matrix is shown as a heat map with colors ranging from red/hot (magnitude = 0) to blue/cold (magnitude = 1). Among all 49 hand gestures, five common gestures for prosthetic

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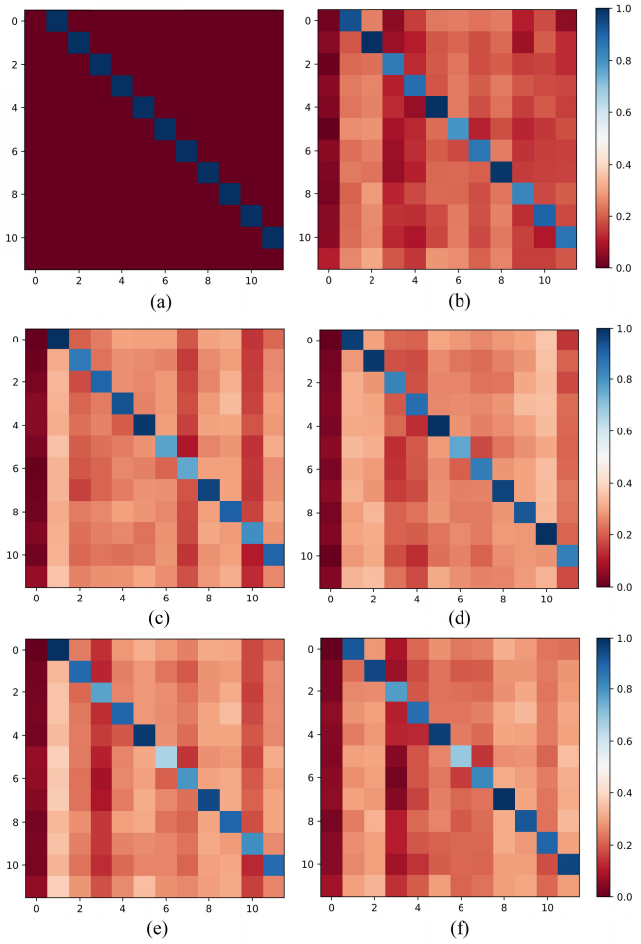


Fig. 5. (a) Shared topology prior. (b)–(f) Refined topologies of different hand gestures ("abduction of all fingers," "pointing index," "wrist flexion," "wrist extension," "wrist pronation," respectively. Each grid heat map illustrates magnitude as color ranging from red/hot (value 0) to blue/cold (value 1). A higher value represents the stronger correlation or connectivity between two nodes. All the gestures are from the tests with Subject No. 1.

hand users ("abduction of all fingers," "pointing index," "wrist flexion," "wrist extension," and "wrist pronation") are selected to show the learned topologies constructed from 12 nodes corresponding to 12 sEMG sensors attached to the arm. In the graphs, a cooler (blue) color indicates the stronger correlation between two nodes while a warmer (red) color represents the weaker connectivity between two nodes. Several observations can be made as follows.

1) Although the predefined relationships show a large influence, the refined topologies are clearly different from the predefined topology, in which only the adjacent pairs of sensors are set to be fully correlated. After the learning processes, some implicit relationships across the muscles (adjacent and nonadjacent) underneath 12 sensors are found by our topology-refinement procedure. The learned graphs suggest that correlations exist in any of the two nodes to different degrees.
2) It is evident that different hand gestures exhibit varied learned graphs, implying distinct muscle coordination for specific hand movements. The learned distinctions contribute to the effective classification of different gestures.

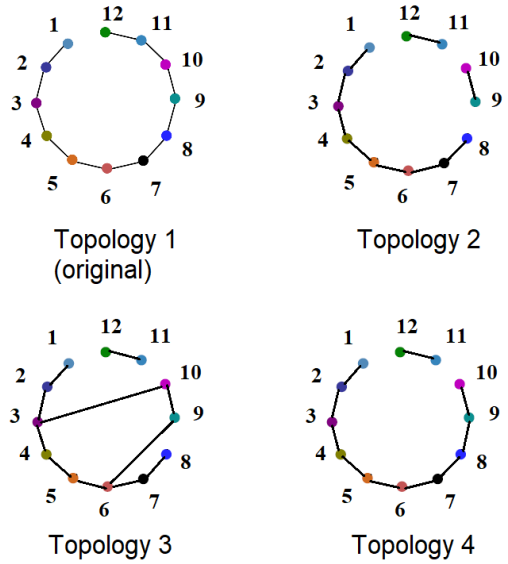


Fig. 6. Tested predefined topologies.

3) For each learned topology, the correlations of the adjacent pairs of sensors are much stronger than the nonadjacent pairs. Besides the impact of the predefined topology, We postulate that the cross-talks between the neighboring sensors may also contribute to the strong correlations between the nodes next to each other.

D. Ablation Studies With NinaPro DB2 Dataset

1) Effectiveness of Predefined Topology: To investigate the impact of a predefined topology, we compare the performance of our CovGCN model with four different predefined topologies, respectively. As shown in Fig. 6, the original topology is the one illustrated in Fig. 1 while the other three topologies have a slight difference between each other. Fig. 3 describes the locations of all the sEMG sensors. Sensor Nos. 1–8 are distributed around the forearm circumferentially at the height of the radio-humeral joint. Therefore, they are always connected together in a predefined topology. Nodes 9 and 10 are jointed together because these two sensors are located at the same height between the radio-humeral joint and the wrist. A similar setting applies to Nodes 11 and 12 since these sensors are both on the upper arm instead of the forearm.

Fig. 7 shows the prior and learned adjacency matrices for each predefined topology, with the same subject conducting the same hand gesture ("abduction of all fingers"). Similar to our observations in Section IV-C, all the learned adjacency matrices are obviously distinct from their prior counterparts, suggesting the efficacy of the topology refinement process. The predefined topology has a clear impact on the learned graph in each case. However, it is difficult to identify clear similarities between the learned adjacency matrices, implying that revealing the underlying muscle synergies thoroughly by sEMG only is a challenging task.

The comparison of classification accuracy using different predefined topologies is shown in Table I. Although there are differences in the learned graphs, all the predefined topologies

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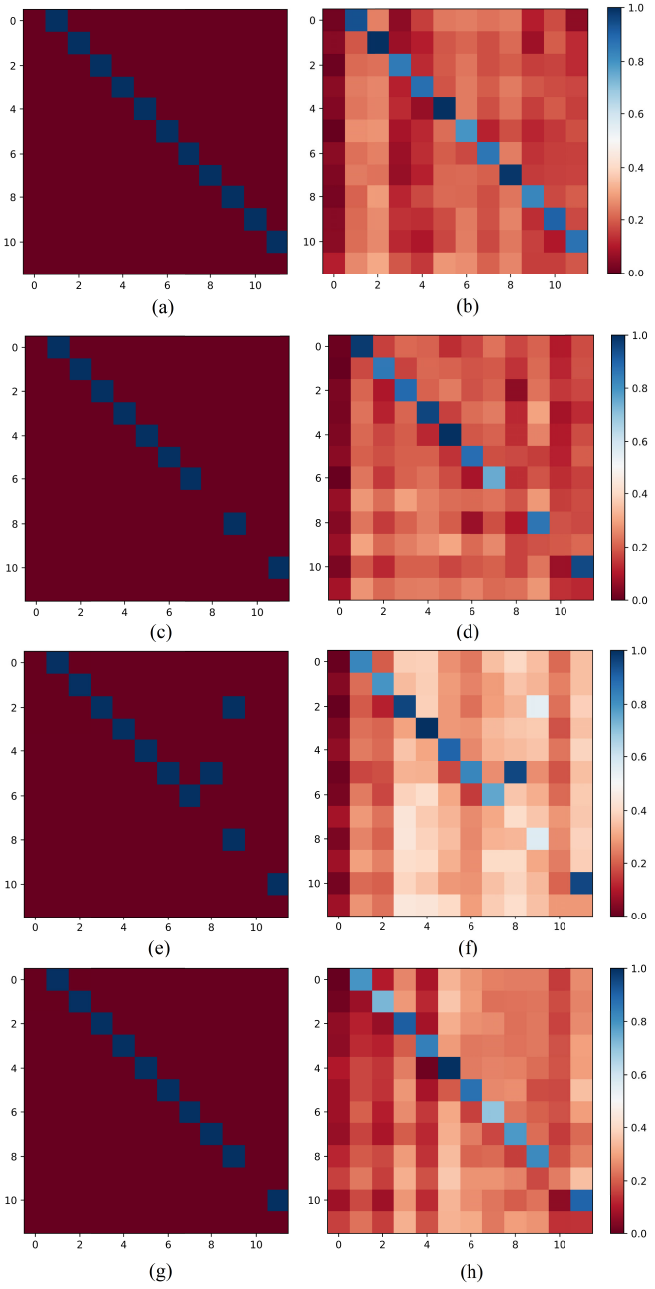


Fig. 7. (a) and (b) Shared topology prior and the refined topology for Topology 1. (c) and (d) Shared topology prior and the refined topology for Topology 2. (e) and (f) Shared topology prior and the refined topology for Topology 3. (g) and (h) Shared topology prior and the refined topology for Topology 4. All the refined topologies are from the same gesture, "abduction of all fingers" of Subject No. 1.

result in a good classification of hand gestures. Among four cases, Topology 1 leads to the worst (relatively) classification accuracy for each exercise. For the other three topologies, their performances are quite close to each other, with the respective advantages for the datasets of the three exercises. For Exercise B, Topology 3 works slightly better than the others. Topology 4 is the best performer for Exercise C while Topology 1 results in the highest classification accuracy for Exercise D. Overall, Topology 1 (the original one) provides consistently good classification for all the exercises. Therefore, it is selected as the predefined topology for all the subsequent experiments.

TABLE I
COMPARISON OF CLASSIFICATION ACCURACY WITH DIFFERENT PREDEFINED TOPOLOGIES

	Topology 1	Topology 2	Topology 3	Topology 4
Exercise B	88.02%	86.14%	**88.38%**	88.05%
Exercise C	80.03%	79.34%	79.87%	**80.10%**
Exercise D	**93.71%**	92.21%	93.26%	92.84%

TABLE II
COMPARISON OF CLASSIFICATION ACCURACY WITH DIFFERENT NUMBER OF GCBS (USING THE DATA IN EXERCISE B)

Method	#Params (M)	GFLOPS	Inference time[1] (ms/2D signal)	Accuracy (%)
Baseline [4]	1.42	1.97	35.1	83.41
5 layers	0.35	1.37	21.7	77.22
7 layers	1.07	2.29	29.6	**88.02**
10 layers	1.52	2.54	42.6	84.45
15 layers	9.35	10.84	65.5	82.14

[1] The experiments are conducted on a computer with Intel Core i7-7700 3.6 GHz processor and NVIDIA GeForce GTX 1080 graphics card.

2) Effectiveness of the Proposed GCB: To investigate the effectiveness of adding the proposed GCB, we gradually increase the number of graph layers from 5 to 15 when conducting the experiments with the data in Exercise B. A model with topology refinement graph convolution (CTR-GC) reported in [4] was used as the baseline in our experiment. The average classification accuracy on 40 subjects in Exercise B reaches 83.41%, which already outperforms the results by the classical ML classifiers (e.g., LDA, KNN, and SVM) and the reported CNN methods in [28] and [29] (using the same database). This indicates the efficacy of CovGCN on this kind of bio-data is strongly influenced by the underlying muscle synergies.

Table II shows the size of the proposed model, the computational complexity in terms of giga floating-point operations per second (GFLOPS), and the processing speed (or latency) in ms per 2-D signal, for different numbers of GCBs. The CovTRM-based model's classification accuracy first rises and then drops, as the number of graph layers increases. The best classification accuracy (88.02%) is achieved using seven graph layers. This model with seven graph layers has fewer parameters than the baseline model reported in [4]. Overall, the best performer has a model size of 1.07 million parameters, requires 2.29 GFLOPS for inference, and takes only 29.6 ms to process a signal. Note that devices such as NVIDIA Jetson Orin can handle up to 2750 GFLOPS, and Raspberry Pi 5 can achieve 31.4 GFLOPS. The above results indicate that the proposed model is suitable for real-time prosthetic hand control with modern portable microprocessors.

It is noteworthy that the optimized graph layer number works for 39 subjects. For only one subject ("S8"), the best performance is obtained by using ten layers, and the difference in accuracy is small between the two configurations (95.83% versus 95.15%). This confirms the effectiveness and stability of the GCB number across different subjects.

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TABLE III
COMPARISON OF CLASSIFICATION ACCURACY WITH DIFFERENT TRMS

Method	TRM 1 [4]	TRM 2 [4]	CovTRM
Accuracy (%)	84.19	85.08	**88.02**

TABLE IV
CLASSIFICATION ACCURACY OF THE CovGCN MODEL (WITH SEVEN GCBS) EVALUATED ACROSS VARIOUS BATCH SIZES

Dataset	Batch size				
	32	**64**	**128**	**256**	**512**
DB-B	86.1%	**87.5%**	86.4%	87.1%	84.9%
DB-C	81.0%	81.3%	**81.9%**	80.8%	80.5%
DB-D	92.8%	93.3%	**93.5%**	92.9%	92.3%

3) Effectiveness of Different TRMs: To investigate the effectiveness of the proposed CovTRM, we compare its performance with two other topology refinement modules (TRMs) reported in [4] when conducting the experiments with the data from 40 subjects in Exercise B. The TRMs in [4] use different correlation modeling functions (pair-wise subtraction and pair-wise concatenation, respectively) to learn specific relationships between the nodes. For comparison, we simply replace the CovTRM with one of the other TRMs for the spatial modeling and keep the temporal modeling structure unchanged in a GCB, as shown in Fig. 2(b). For each set of experiments with different TRMs, the same number of graph layers (seven GCBs) are employed. Table III shows the accuracy of HGR with the GCN architectures based on our CovTRM and the other two TRMs, respectively. It is noted that all the tested GCN architectures achieve better classification accuracy than that of the baseline model [4] (shown in Table II). Among all the tested GCN architectures with different TRMs, the one based on CovTRM clearly outperforms the other two in terms of classification accuracy. The attention mechanism can effectively find the semantic information to refine the input covariance and subsequently provide a better way to identify correlations between different channels of muscular signals for spatial modeling.

4) Effectiveness of Batch Size: In this ablation study, we analyze the impact of batch size on the performance of the proposed CovGCN. To this end, we trained and evaluated the CovGCN with seven GCBs across five different batch sizes: 32, 64, 128, 256, and 512. Table IV shows that, for the dataset DB-B, a batch size of 64 achieves the highest classification accuracy of 87.5%. For the datasets DB-C and DB-C, the optimal batch size is 128 with accuracies of 81.9% and 93.5%, respectively. Batch sizes between 64 and 128 produce acceptable classification performances across three datasets.

The experimental results also indicate that a larger batch size of 512 leads to a slight decrease in accuracy for all examined datasets. A possible explanation for this finding is that, although the large batch size can stabilize gradients during backpropagation, it may also decrease the noise in gradient updates. This causes the model to get stuck in the local minima of the loss surface, which reduces the generalization ability of the model.

TABLE V
COMPARISON OF CLASSIFICATION ACCURACY WITH OTHER ML/DL METHODS

Method	Dataset	Accuracy(%)
KNN [26]	DB-B	75.17
LDA [26]	DB-B	79.95
SVM [26]	DB-B	79.50
SVM [28]	DB-B	81.07
MLP [26]	DB-B	80.97
CNN [26]	DB-B	82.87
CNN [28]	DB-B	82.22
CNN [29]	DB-B	83.79
MSFEnet [27]	DB-B	86.21
Transformer [14]	DB-B	88.91
CovGCN (ours)	DB-B	**88.02**
SVM [28]	DB-C	71.08
CNN [28]	DB-C	72.62
Transformer [14]	DB-C	81.44
CovGCN (ours)	DB-C	**81.30**
SVM [28]	DB-D	88.56
CNN [28]	DB-D	89.54
MSFEnet [27]	DB-D	92.53
Transformer [14]	DB-D	93.84
CovGCN (ours)	DB-D	**93.30**

E. Comparison With Other ML and DL Methods Using NinaPro DB2 Dataset

Table V shows the comparison of classification accuracy between the proposed CovGCN and some other recently reported ML/DL methods using the same NinaPro datasets. For consistency, all the compared results are based on the window segmentation of 200 ms. From Table V, it is seen that our proposed CovTRM-based model outperforms most of the other ML/DL methods for all three exercises of hand movements. Compared to the Transformer model [14], the proposed model has slightly lower classification accuracy. Compared to the CNN models, our CovGCN model is much better, indicating its advantage in sEMG-based human body action. In typical GCN applications, the physical links of body parts are explicit. In contrast, in sEMG-based human body action recognition, the key challenge of applying GCN is to identify topologies due to the implicit and complicated gesture-dependent muscle synergies. Our proposed CovTRM shows the efficacy in learning "unseeable" muscle networks for topology refinement of GCN, leading to superior spatial modeling and consequently the marked higher classification accuracy than the CNN-based methods.

From Table V, it is seen that the classification accuracy by our proposed GCN model is higher with the DB-D dataset than the DB-B and DB-C datasets. This is due to the relative simplicity of hand gestures in Exercise D (mainly single-digit or double-digit movements). Besides, the number of gestures is also the fewest. By contrast, the pattern recognition accuracy with the DB-C is the lowest among all three datasets because it includes more hand gestures than the other two. Besides,

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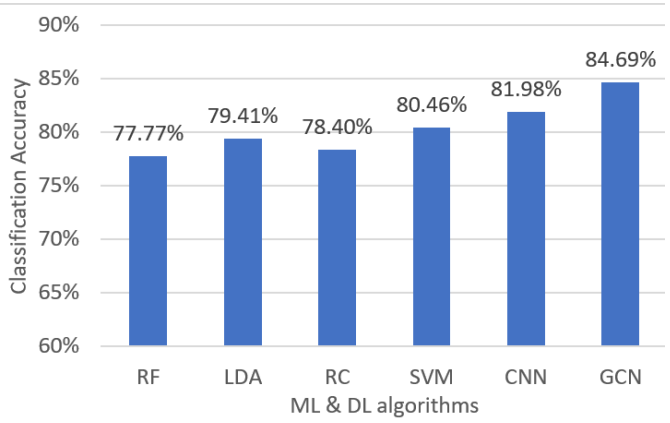


Fig. 8. HGR accuracy with different algorithms for UOW Dataset (variable limb position).

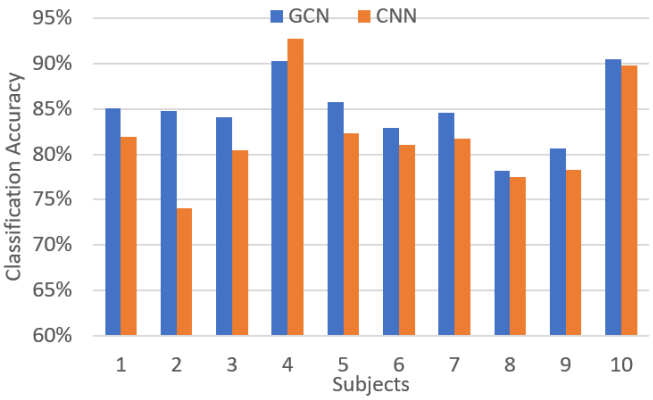


Fig. 9. CovGCN versus CNN, for all the subjects from UOW Dataset (variable limb position).

some gestures in DB-C are similar to each other, e.g., "Large diameter grasp," "Small diameter grasp," "Medium wrap," "Power sphere grasp," "Precision sphere grasp," "Prismatic pinch grasp," and "Tip pinch grasp," making it challenging to classify those pairs/triad. This observation of classification difference among three datasets is consistent with other reported studies in Table V. A practical solution to this problem in prosthetic hand applications is to adopt underactuation or use built-in sensors. This will enable the hand to comply with different object profiles when the same hand gesture is activated [31]. Accordingly, similar hand gestures such as different types of sphere grasps can be grouped into one category for classification.

F. Comparison With CNN Method on Handling Arm Positions Using UOW Dataset

In the NinaPro DB2 Dataset, all the hand gestures were performed with a fixed limb position for each subject. To explore the HGR performance of our GCN model with the impact of variable limb position, we conducted additional experiments and analysis using the sEMG data from the UOW Dataset. Fig. 8 shows the comparison of classification accuracy (averaged across ten subjects) using the same data but different ML/DL algorithms. The DL methods outperform the traditional ML methods, providing better classification when the limb position varies. Between the two DL methods, our proposed GCN model clearly shows higher HGR accuracy than that of the CNN model reported in [8] (84.69% versus 81.98%), indicating its superior resilience to the limb position effects.

Fig. 9 shows the comparison of classification accuracy between the GCN model and the CNN mode for each subject. Except for Subject #4, the GCN model outperforms the CNN model for all the subjects, demonstrating its consistent efficacy and advantage in overcoming the limb position effects across different people.

G. Discussion on Limitations

The proposed GCN has shown several benefits, but it can still be improved in a number of aspects. *First*, for consistent comparisons with the other reported models, we have employed the Ninapro DB2 dataset and UOW dataset to test the proposed GCN model. However, both datasets include intact subjects only. Intact subjects are commonly used to evaluate the performance of sEMG-based HGR algorithms in the research literature, but differences exist in muscle activations between intact and amputated arms. Therefore, validation with the sEMG data from amputee subjects would provide more insights for the specific HGR application in prosthetic hand control. *Second*, in this article we have evaluated different methods systematically via offline analysis. However, further evaluations through online tests would be useful to establish the feasibility of the proposed technology for the end-users. *Third*, this work primarily focuses on proposing a novel graph-based model and evaluating its classification performance in subject-specific scenarios. We acknowledge that this study could be extended by including a cross-subject setting to evaluate the CovGCN's ability to generalize for data from unseen subjects. We leave the exploration of more extensive cross-subject validation to a future study.

V. Conclusion

In this study, a novel CovTRM-integrated GCN architecture is proposed for sEMG-based HGR, which can consequently offer a robust HMI for reliable prosthetic hand control. The performance of the proposed DL method is tested on Ninapro DB2, a publicly available HRG database. The proposed TRM can effectively learn the relationships of different nodes/sensors corresponding to diverse hand gestures. The refined GCN graphs indicate the underlying muscle networks/synergies accordingly. The GCN model is optimized in terms of predefined topology, graph layers, and TRM functions. The classification results show that the proposed GCN model achieves state-of-the-art robustness in sEMG-based HGR and outperforms most of the DL models which are tested on the same database. The results also indicate that GCN is clearly more advantageous than CNN in spatial modeling of sEMG-based human body actions. Furthermore, our proposed GCN model is tested on the UOW Dataset to investigate its HGR performance in the presence of one major confounding factor in practice: variable limb position. The results show that the proposed GCN model performs better than all the

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traditional ML methods as well as the CNN model with regard to overcoming the impact of limb position on HGR robustness, which can greatly improve the commercial and clinical viability of pattern recognition-based myoelectric hand control.

Future directions include developing a real-time prosthetic hand control system with a portable computer such as NVIDIA Jetson Orin, and investigating online performance (e.g., completion rate, completion time, and real-time accuracy) with disabled end-users. Moreover, the proposed GCN model could be extended to process inputs from more multimodal sensors and enhance its accuracy and robustness. Another venue for further research involves investigating cross-subject validation in conjunction with transfer learning to accelerate training and increase robustness for unseen subjects.

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